Mattia D'Urso

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Citizenship: Italian (EU Citizen) | **Languages:** Italian (Native), English (C1 - Full Professional)

Education

Feb 2024 – Today — Ph.D. in **Computer Science and AI**
at Institute of Visual Computing, Graz University of Technology (Austria)
- Funded by Sony Europe and under the supervision of Prof. Friedrich Fraundorfer.

Topics: 3D Computer Vision · Deep Learning · Structure-from-Motion · Geometric Optimization

2020 – 2022 — M.Sc. in **Computer Science and AI** *at the Universities of Klagenfurt and Udine*

2017 – 2020 — B.Sc. in **Computer Science** *at University of Udine*

Experiences

May 2026 – May 2027 — **ETH Zürich**, *Zürich, Visiting Researcher*
- Supervised by Prof. Marc Pollefeys and Dr. Dániel Baráth at the CVG group.
- Extending 3D foundation models capabilities for real-world deployment.

Jun 2023 – Aug 2023 — **Canva**, *Vienna, Applied Scientist Intern*
- Adapted Discrete Stable Diffusion paper for Layout Generation on a custom dataset.
- Improved by 20% the number of perfect samples generated by enhancing data and model.

Mar 2022 – Nov 2022 — **Infineon Technologies AG**, *Villach, Applied Scientist Intern*
- Developed a predictive maintenance system based on time series using Deep Learning models.

Publications

Boosting 3D Foundation Models with Edge-based Pose Optimization
Mattia D'Urso, C. Sormann, M. Rossi, F. Fraundorfer
European Conference of Computer Vision (ECCV) 2026
- *Trackless refinement of cameras and geometry starting from 3D foundation models initializations, matching Bundle Adjustment accuracy with up to 98% lower runtime. 110+ Stars on GitHub.*

TerraSky3D: Multi-View Reconstructions of European Landmarks in 4K
Mattia D'Urso, Y. Hu, C. Sormann, M. Rossi, F. Fraundorfer
3DMV @ CVPRW 2026 **[Best Paper Award (Runner-Up)]**
- *4K dataset comprising 150 European landmarks (50K images) with curated poses and depth maps, bridging ground-level and aerial views for 3D reconstruction. 7K+ downloads on Hugging Face.*

A Streamlined Attention-based Network for Descriptor Extraction
Mattia D'Urso, E. Santellani, C. Sormann, M. Rossi, A. Kuhn, F. Fraundorfer
13th *International Conference on 3D Vision (3DV) 2026*
- *Descriptor module trainable on top of any keypoint detector, significantly improving pose estimation on high-resolution images.*

A LoD of Gaussians: Out-of-Core Training and Rendering for Seamless Ultra-Large Scene
F. Windisch, T. Köhler, L. Radl, Mattia D'Urso, M. Steiner, D. Schmalstieg, M. Steinberger. *SIGGRAPH 2026*

PyTorchGeoNodes: Enabling Differentiable Shape Programs for 3D Shape Reconstruction
S. Stekovic, A. Artykov, S. Ainetter, Mattia D'Urso, F. Fraundorfer. *CVPR 2025*

Other Experiences

2026 – Today — **Open Source @ Kornia — Geometric Computer Vision Library for Spatial AI**
- Expanding Kornia's evaluation toolkit with camera pose related metrics.

2025 – Today — **Mentor @ Lead The Future**
- Invited as a mentor within LTF, a selective STEM community (15% acceptance).